

Emily Root

Game Developer, Gameplay and UI Programming

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SUMMARY

Game developer who has built games in Unity and Godot, and a year of building production software for paying clients. I design systems that teach players: block-based coding puzzles, financial literacy mechanics, and AI opponents.

GAME PROJECTS

Gainz: The Financial Literacy Game

Godot, GDScript | 2024 | [Play on itch.io](#)

- Team lead and programmer on a 5-person team building a financial literacy game for Katabasis, an educational nonprofit serving under-resourced youth
- Designed core loops that turn financial choices into short, replayable decisions with visible consequences

Resource Rush

Unity, C# | 2023 (NC State Game2Learn Lab)

- Ported an educational farming game from JavaScript to Unity, then added new features and art
- Built an in-game block-based coding system where K-12 players program farm tasks to solve puzzles

Game jams (GMTK Game Jam)

2021 – 2024

- [Pollenation](#) (2024): main programmer on a team of 4; beehive simulation played across resource tiles
- [Cat's Cradle](#) (2023): programmer on a team of 6; tile-based puzzle game built in 48 hours
- [Dice Warp](#) (2022): solo; Unity tile puzzle where the die must reach the goal on the correct face
- [God's Year Off](#) (2021): solo art and code; Asteroids-style pixel-art game, my first published title

More projects

- [Squidnapped](#) (PuzzleScript, 2023): solo puzzle game about a squid escaping an aquarium by blocking cameras
- emilyroot.com: interactive 3D portfolio built in Three.js with Claude Code and Claude Design

EXPERIENCE

Software Engineer, BrightVine Solutions

Remote | July 2025 – Present

- Own a production TypeScript/Angular app from first commit to release, used by 10+ large institutional clients
- Build complex, state-heavy UI with multi-step forms, modals, and live data for non-technical users
- Find and fix hard bugs in live environments, such as state lost on failed saves and silent data drops, and work directly with users to reproduce issues
- Run releases, change logs, and patch notes on a steady shipping cadence

Educational Game Developer, NC State Game2Learn Lab

Raleigh, NC | Jun – Aug 2023

- Ported Resource Rush, a K-12 coding game, from JavaScript to Unity (C#)
- Built an in-game block-based coding system that drives farm actions and turns tasks into puzzles

Teaching Assistant, NC State University

Raleigh, NC | Jan 2022 – May 2025

- Taught programming fundamentals in Python and Snap to 200+ students across seven semesters
- Promoted to lead TA; used what I saw beginners struggle with to improve course materials, much like playtesting

SKILLS

- **Engines and tools:** Unity, Godot, Unreal Engine, GameMaker Studio 2, PuzzleScript, Three.js, Blender, Live2D, Git
- **Languages:** C#, C++, C, Python, Java, GDScript, TypeScript, JavaScript, SQL
- **Game systems:** gameplay programming, UI/UX, game AI (search, decision-making), educational game design

EDUCATION

B.S. in Computer Science, North Carolina State University

Raleigh, NC | May 2025

Game coursework: Game Engine Foundations, Interactive Game Design, Building Game AI, Computational Visual Narrative, Intro to AI, Operating Systems

LEADERSHIP AND VOLUNTEERING

Officer, STARS Computing Corps

Aug 2021 – May 2024

- Taught game development in Unity, Unreal Engine, and JavaScript to 50+ middle and high school students